

READING PASSAGE 3

You should spend about 20 minutes on Questions 27–40, which are based on Reading Passage 3 below.

Can hurricanes be moderated or diverted

A

Each year, massive swirling storms bringing along winds greater than 74 miles per hour wipe across tropical oceans and land on shorelines—usually devastating vast swaths of territory. When these roiling tempests strike densely inhabited territories, they have the power to kill thousands and cause property damage worth of billions of dollars. Besides, absolutely nothing stands in their way. But can we ever find a way to control these formidable forces of nature?

B

To see why hurricanes and other severe tropical storms may be susceptible to human intervention, a researcher must first learn about their nature and origins. Hurricanes grow in the form of thunderstorm clusters above the tropical seas. Oceans in low-latitude areas never stop giving out heat and moisture to the atmosphere, which brings about warm, wet air above the sea surface. When this kind of air rises, the water vapour in it condenses to form clouds and precipitation. Condensation gives out heat in the process the solar heat is used to evaporate the water at the ocean surface. This so-called invisible heat of condensation makes the air more buoyant, leading to it ascending higher while reinforcing itself in the feedback process. At last, the tropical depression starts to form and grow stronger, creating the familiar eye — the calm centre hub that a hurricane spins around. When reaching the land, the hurricane no longer has a continuous supply of warm water, which causes it to swiftly weaken.

C

Our current studies are inspired by my past intuition when I was learning about chaos theory 30 years ago. The reason why long-range forecasting is complicated is that the atmosphere is highly sensitive to small influences and tiny mistakes can compound fast in the weather-forecasting models. However, this sensitivity also made me realise a possibility: if we intentionally applied some slight inputs to a hurricane, we might create a strong influence that could affect the storms, either by steering them away from densely populated areas or by slowing them down. Back then, I was not able to test my ideas, but thanks to the advancement of computer simulation and remote-sensing technologies over the last 10 years, I can now renew my enthusiasm in large-scale weather control.

D

To find out whether the sensitivity of the atmospheric system could be exploited to adjust such robust atmospheric phenomena as hurricanes, our research team ran simulation experiments on computers for a hurricane named Iniki that occurred in 1992. The current forecasting technologies were far from perfect, so it took us by surprise that our first simulation turned out to be an immediate success. With the goal of altering the path of Iniki in mind, we first picked the spot where we wanted the storm to stop after six hours. Then we used this target to generate artificial observations and put these into the computer model.

E

The most significant alteration turned out to be the initial temperatures and winds. Usually, the temperature changes across the grid were only tenths of a degree, but the most noteworthy change, which was an increase of almost two degrees Celsius, took place in the lowest model layer to the west of the storm centre. The calculations produced wind-speed changes of two or three miles per hour. However, in several spots, the rates shifted by as much as 20 mph due to minor redirections of the winds close to the storm's centre. In terms of structure, the initial and altered versions of Hurricane Iniki seemed almost the same, but the changes in critical variables were so substantial that the latter one went off the track to the west during the first six hours of the simulation and then travelled due north, leaving Kauai untouched.

F

Future earth-orbiting solar power stations, equipped with large mirrors to focus the sun's rays and panels of photovoltaic cells to gather and send energy to the Earth, might be adapted to beam microwaves which turn to be absorbed by water vapour molecules inside or around the storm. The microwaves would cause the water molecules to vibrate and heat up the surrounding air, which then leads to the hurricane slowing down or moving in a preferred direction.

G

Simulations of hurricanes conducted on a computer have implied that by changing the precipitation, evaporation and air temperature, we could make a difference to a storm's route or abate its winds. Intervention could be in many different forms: exquisitely targeted clouds bearing silver iodide or other rainfall-inducing elements might deprive a hurricane of the water it needs to grow and multiply from its formidable eyewall, which is the essential characteristic of a severe tropical storm.

Questions 27-30

Reading Passage 3 has seven paragraphs, A-G.

Choose the correct heading for each paragraph from the list of headings below.

Write the correct number, i-viii, in boxes 27-33 on your answer sheet.

Lists of headings

- i** Hurricanes in history.
- ii** How hurricanes form.
- iii** How a laboratory exercise re-route a hurricane.
- iv** Exciting ways to utilise future technologies.
- v** Are hurricanes unbeatable?
- vi** Re-visit earlier ideas.
- vii** How lives might have been saved.
- viii** A range of low-tech methods.

27 Paragraph **A**

28 Paragraph **B**

29 Paragraph **C**

30 Paragraph **D**

31 Paragraph **E**

32 Paragraph **F**

33 Paragraph **G**

Questions 34-38

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer. Write your answers in boxes **34-38** on your answer sheet.

Hurricanes originate as groups of **34**..... over the tropical oceans. Low-latitude seas continuously provide heat and moisture to the atmosphere, producing warm, humid air above the sea surface. When this air rises, the water vapour in it condenses to form clouds and precipitation. **35**..... releases heat—the solar heat it took to evaporate the water at the ocean surface. This so-called latent **36** of condensation makes the air more buoyant, causing it to ascend still higher in a self-reinforcing feedback process. Eventually, the tropical depression begins to organise and strengthen, forming the familiar **37**..... the calm central hub around which a hurricane spins. On passing over **38**..... the hurricane's sustaining source of warm water is cut off, which leads to the storm's rapid weakening.

Questions 39-40

Choose the correct letter, A, B, C or D.

Write the correct letter in boxes 39 and 40 on your answer sheet.

39. What encouraged the writer to restart researching hurricane control?

- A** The huge damage hurricane triggers
- B** The developments in computer technologies
- C** The requirement of some local people
- D** The chaos theory learnt as a student

40. What was the writer's reaction after their first experiment?

- A** Surprised that their intervention had not achieve a lot.
- B** Ecstatic with the achievement the first experiment had
- C** Surprised that their intervention had the intended effect
- D** Regretful about the impending success.

Answers

- 27. v
- 28. ii
- 29. vi
- 30. iii
- 31. vii
- 32. iv
- 33. viii
- 34. thunderstorms
- 35. Condensation
- 36. heat
- 37. eye
- 38. land
- 39. B
- 40. C

